The Data and Existing Methodology

This project will use game film from the Pomona-Pitzer Women's Water Polo 2025-2026 season. Pomona-Pitzer Water Polo is Division III, however, plates against Division I, II, and III teams as well. Because opponents widely vary in difficulty, only games played against ranked opponents will be used in this analysis. I will use the top 25 rankings made by the National Collegiate Athletic Association (NCAA) and the Collegiate Water Polo Association (CWPA). The NCAA uses metrics like the Rating Percentage Index (RPI) or selection criteria to evaluate win-loss records, schedule strength, and championship qualification likeliness to create rankings. The CWPA publishes weekly top 25 rankings voted on by coaches. Both associations' rankings are respected and used by the varsity water polo community, which is why I chose to reference both. If one association ranks a team in the top 25 that the other association did not, I will include it. Although they are not ranked in the top 25 by either association, I chose to include Claremont-Mudd-Scripps, as they are PP's rivals and PPWWP's biggest Division III opponent. PPWWP played 17 games against a ranked opponent or CMS. Limiting the sample in this way allows me to focus on games played against relatively strong and competitively important opponents while keeping the amount of film that must be processed manageable.

The game film is stored and tagged using Hudl Sportscode. During or after each game, a coach or student-athlete will tag the game meaning mark certain actions such as shots, turnovers, exclusions, and goals in the game. However, on top of the tagging process, a coach may select each shot, for example, then add additional information. This process takes a lot of time. The tagged game is able to be exported, where the CSV file shows the tagged action, the start time, and end time. The unit of observation in the finland dataset will be a shot, where each row represents a Pomona-Pitzer Women's Water polo shot. The columns will describe characteristics of the shot. Some of these characteristics are already in the Hudl Sportscode and other characteristics will need to be coded. The resulting dataset will be standardized enough that it can be reproduced across games and potentially reused for future seasons.